



Module Presenter's Manual *for* **Building Modern Websites**

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Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	March 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Explain the evolution of HTML
- Explain the page structure used by HTML
- List the drawbacks in HTML 4 and XHTML
- List the new features of HTML5
- Explain CSS and JavaScript
- Explain jQuery
- Explain browser support for HTML5
- Explain the elements constituting an HTML tag
- Describe doctype
- Explain the basic tags in HTML
- Describe container and standalone tags
- Explain the role of HTML5 in Mobile devices
- Explain the Heading tag
- Explain the different tags related to formatting
- Explain monospaced font, preformatted text, and block quotation
- Describe the different types of lists
- Explain the procedure to change the background color and image
- Describe hyperlinks
- Explain how to hyperlink to a Web page and e-mail address
- Explain how to hyperlink to anchors and other content
- Identify the new functions of CSS3
- Explain the different types of selectors
- Explain nested tags
- Define Classes and IDs for applying styles
- Explain the process to apply styles to hyperlink
- List and explain text and font styles
- Explain paragraph indentation and application of border
- Explain horizontal paragraph alignment
- Explain vertical spacing within a paragraph
- Explain graphic formatting in Web pages
- Explain graphic insertion, sizing, and padding
- Explain CSS3 Animation
- Describe the use of CSS3 on Mobile devices
- Describe flexbox and grid layouts
- Describe different flexbox properties
- Explain different grid properties
- Explain HTML5 semantic tags
- Explain HTML5 semantic tag layouts
- Explain the usage of navigation bar
- Describe a text-based and graphical navigation bar
- Explain image mapping
- Explain divisions in HTML5
- Describe HTML5 forms
- Explain the working of new input types in HTML5
- Explain the new Form attributes
- Explain the new Form elements
- Define hidden fields
- Describe how to create and format tables

- Explain the table size and the width of a column
- Explain the process of merging table cells
- Explain the page layout for tables
- Describe the need for multimedia in HTML5
- List the supported media types in HTML5
- Explain the audio elements in HTML5
- Explain the video elements in HTML5
- Explain the accessibility of audio and video elements
- Describe how to deal with non-supporting browsers
- Explain scripting
- Explain the JavaScript language
- Explain the client-side and server-side JavaScript
- List the variables and data types in JavaScript
- Describe the JavaScript methods to display information
- Explain escape sequences and built in functions in JavaScript
- Explain events and event handling
- Explain jQuery
- Describe how to use the jQuery Mobile
- Explain operators and their types in JavaScript
- Explain regular expressions in JavaScript
- Explain decision-making statements in JavaScript
- Explain break and continue statement
- Explain single-dimensional arrays
- Explain multi-dimensional arrays
- Explain for..in loop
- Explain JSON
- Describe Canvas in HTML5
- Explain the procedure to draw lines
- Explain the procedure to use color and transparency
- Explain the procedure to work with various drawing objects, images, and text
- Describe the procedure to create Web page events with JavaScript and jQuery
- Describe the process of including external content in Web pages
- Explain Web storage in HTML5
- Explain session storage and local storage
- Explain Indexed DB API
- Describe a native app
- Explain difference between native apps and HTML5 apps
- Describe advantages of native and HTML5 apps
- List steps to convert HTML5 apps to native apps
- Explain geolocation and its use in HTML5
- Explain the Google Maps API
- Explain the drag-and-drop operations in HTML5
- Explain the concept of Application Cache
- Describe features of different mobile devices
- List different types of platforms available for mobile devices
- Explain design and architectural aspects of a mobile Website
- Explain the requirements for developing and testing of a mobile Website
- Explain HTML5 support for a mobile Website
- List the best practices for optimizing a mobile Website
- Explain the use of HTML5, CSS, and JavaScript in a real-world application

2. Information on Session Allocation

Module	Online (No. of Hours)	Self-study (No. of Hours)
HTML5	40	12

Throughout this Presenter's Manual, the module **Building Modern Websites** will be referred to as **HTML5**.

3. Week-wise Session Schedule

- A session is of 2 hours duration.

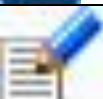
➤ Week-Wise Schedule

Week	Day 1	Day 2	Day 3	Day 4	
1	Session 1 HTML5 – TL1	Session 2 HTML5 – TL2	Session 3 HTML5 – TL3	Session 4 HTML5 – TL4	S1- S2
2	Session 5 HTML5 – TL5	Session 6 HTML5 – TL6	Session 7 HTML5 – TL7	Session 8 HTML5 – TL8	
3	Session 9 HTML5 – TL9	Session 10 HTML5 – TL10	Session 11 HTML5 – TL11	Session 12 HTML5 – TL12	S3- S4
4	Session 13 HTML5 – TL13	Session 14 HTML5 – TL14	Session 15 HTML5 – TL15	Session 16 HTML5 – TL16	
5	Session 17 HTML5 – TL17	Session 18 HTML5 – TL18	Session 19 HTML5 – TL19	Session 20 HTML5 – TL20	S5- S6

TL: Online Session

S: Self-study Hours (over and above the module duration)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Show Me How - Student can view a step-wise simulation/demonstration of the module related topics.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	HTML5 – TL1	All the topics as listed below from Session 1 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 1 – Introduction to the Web</u> ➤ Explain the evolution of HTML ➤ Explain the page structure used by HTML ➤ List the drawbacks in HTML 4 and XHTML ➤ List the new features of HTML5 ➤ Explain CSS ➤ Explain JavaScript ➤ Explain jQuery ➤ Explain browser support for HTML5	<u>FrontEnd Technologies for Beginners</u> SG – Session 1 XP – Session 1 TG – Session 1
2	HTML5 – TL2	All the topics as listed below from Session 2 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 2 – Introduction to HTML5</u> ➤ Explain the elements constituting an HTML tag ➤ Describe doctype ➤ Explain the basic tags in HTML ➤ List the different data types, attributes, and entities of HTML5 ➤ Describe container and standalone tags ➤ Explain the role of HTML5 in Mobile devices	<u>FrontEnd Technologies for Beginners</u> SG – Session 2 XP – Session 2 TG – Session 2
3	HTML5 – TL3	All the topics as listed below from Session 3 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 3 – Formatting Text Using Tags</u> ➤ Explain the Heading tag ➤ Explain the different tags	<u>FrontEnd Technologies for Beginners</u> SG – Session 3 XP – Session 3 TG – Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		related to formatting ➤ Explain monospaced font, preformatted text, and block quotation ➤ Describe the different types of lists ➤ Explain the procedure to change the background color and image	
4	HTML5 – TL4	All the Workshop topics from Session 1 to Session 3 of <i>FrontEnd Technologies for Beginners</i> course should be covered in this session.	<u>FrontEnd Technologies for Beginners</u> (OnlineVarsity) Session 1-3
	S1-S2	All the Assignment topics from Session1 to Session 3 of <i>FrontEnd Technologies for Beginners</i> course should be covered in this session.	<u>FrontEnd Technologies for Beginners</u> (OnlineVarsity) Session 1-3
5	HTML5 – TL5	All the topics as listed below from Session 4 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 4 – Creating Hyperlinks and Anchors</u> ➤ Describe hyperlinks ➤ Explain absolute and relative paths ➤ Explain how to hyperlink to a Web page and e-mail address ➤ Explain how to hyperlink to anchors and other content	<u>FrontEnd Technologies for Beginners</u> SG – Session 4 XP – Session 4 TG – Session 4
6	HTML5 – TL6	All the topics as listed below from Session 5 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 5 – Introduction to CSS3</u> ➤ Identify the new functions of CSS3 ➤ Explain the different types of selectors ➤ Explain nested tags ➤ Define Classes and IDs for applying styles ➤ Explain the process to apply	<u>FrontEnd Technologies for Beginners</u> SG – Session 5 XP – Session 5 TG – Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
		styles to hyperlink	
7	HTML5 – TL7	All the topics as listed below from Session 6 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 6 – Formatting Using Style Sheets</u> ➤ List and explain text and font styles ➤ Describe inline spans ➤ Explain paragraph indentation and application of border ➤ Explain horizontal paragraph alignment ➤ Explain vertical spacing within a paragraph	<u>FrontEnd Technologies for Beginners</u> SG – Session 6 XP – Session 6 TG – Session 6
8	HTML5– TL8	All the topics as listed below from Session 7 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 7 – Displaying Graphics and CSS3 Animation</u> ➤ Explain graphic formatting in Web pages ➤ Explain graphic insertion, sizing, and padding ➤ Explain CSS3 Animation ➤ Describe the use of CSS3 on Mobile devices	<u>FrontEnd Technologies for Beginners</u> SG – Session 7 XP – Session 7 TG – Session 7
9	HTML5 – TL9	All the topics as listed below from Session 8 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 8 – Understanding Layouts in CSS3</u> ➤ Describe flexbox and grid layouts ➤ Describe different flexbox properties ➤ Explain different grid properties	<u>FrontEnd Technologies for Beginners</u> SG – Session 8 XP – Session 8 TG – Session 8
10	HTML5 – TL10	All the topics as listed below from Session 9 of <i>FrontEnd Technologies for Beginners</i> book should be	<u>FrontEnd Technologies for Beginners</u>

Session No.	Session Title	Session Details	Deliverables' Mapping
		covered in this session. <u>Session 9 – HTML Layout, Forms, and Elements</u> ➤ Explain HTML5 semantic tags ➤ Explain HTML5 semantic tag layouts ➤ Explain the usage of navigation bar ➤ Describe a text-based and graphical navigation bar ➤ Explain image mapping ➤ Explain divisions in HTML5	SG – Session 9 XP – Session 9 TG – Session 9
11	HTML5 – TL11	All the topics as listed below from Session 9 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 9 – HTML Layout, Forms, and Elements</u> ➤ Describe HTML5 forms ➤ Explain the working of new input types in HTML5 ➤ Explain the new Form attributes ➤ Explain the new Form elements ➤ Define hidden fields	<u>FrontEnd Technologies for Beginners</u> SG – Session 9 XP – Session 9 TG – Session 9
12	HTML5 – TL12	All the Workshop topics from Session 9 of <i>FrontEnd Technologies for Beginners</i> course should be covered in this session.	<u>FrontEnd Technologies for Beginners (OnlineVarsity)</u> Session 9
	S3-S4	All the Assignment topics from Session 4 to Session 9 of <i>FrontEnd Technologies for Beginners</i> course should be covered in this session.	<u>FrontEnd Technologies for Beginners (OnlineVarsity)</u> Session 4-9
13	HTML5 – TL13	All the topics as listed below from Session 10 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 10 – HTML5 Tables, Audio, and Video</u>	<u>FrontEnd Technologies for Beginners</u> SG – Session 10 XP – Session 10 TG – Session 10

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Describe how to create and format tables ➤ Explain the table size and the width of a column ➤ Explain the process of merging table cells ➤ Explain the page layout for tables ➤ Describe the need for multimedia in HTML5 ➤ List the supported media types in HTML5 ➤ Explain the audio elements in HTML5 ➤ Explain the video elements in HTML5 ➤ Explain the accessibility of audio and video elements ➤ Describe how to deal with non-supporting browsers	
14	HTML5 – TL14	All the topics as listed below from Session 11 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 11 – JavaScript – I</u> ➤ Explain scripting ➤ Explain the JavaScript language ➤ Explain the client-side and server-side JavaScript ➤ List the variables and data types in JavaScript ➤ Describe the JavaScript methods to display information ➤ Explain escape sequences and built in functions in JavaScript ➤ Explain events and event handling ➤ Explain jQuery ➤ Describe how to use the jQuery Mobile	<u>FrontEnd Technologies for Beginners</u> SG – Session 11 XP – Session 11 TG – Session 11
15	HTML5 – TL15	All the topics as listed below from Session 11 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 11 – JavaScript – I</u>	<u>FrontEnd Technologies for Beginners</u> SG – Session 11 XP – Session 11 TG – Session 11

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain operators and their types in JavaScript ➤ Explain regular expressions in JavaScript ➤ Explain decision-making statements in JavaScript ➤ Explain while loop ➤ Explain for loop ➤ Explain do..while loop ➤ Explain break and continue statement ➤ Explain single-dimensional arrays ➤ Explain multi-dimensional arrays ➤ Explain for..in loop	
16	HTML5 – TL16	All the topics as listed below from Session 12 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 12 – JavaScript – II</u> ➤ Explain functions ➤ Explain parameterized functions ➤ Explain return statement ➤ Describe objects ➤ Explain different browser objects ➤ Describe DOM and its objects ➤ Identify the use of Promise.any ➤ Explain Private class methods ➤ Explain JSON	<u>FrontEnd Technologies for Beginners</u> SG – Session 12 XP – Session 12 TG – Session 12
17	HTML5 – TL17	All the topics as listed below from Session 13 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 13 – Canvas and Web Storage in HTML5</u> ➤ Describe Canvas in HTML5 ➤ Explain the procedure to draw lines ➤ Explain the procedure to use color and transparency ➤ Explain the procedure to work with various drawing objects, images, and text ➤ Describe the procedure to create Web page events with	<u>FrontEnd Technologies for Beginners</u> SG – Session 13 XP – Session 13 TG – Session 13

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ JavaScript and jQuery ➤ Describe the process of including external content in Web pages ➤ Explain Web storage in HTML5 ➤ Explain session storage and local storage ➤ Explain Indexed DB API ➤ Describe a native app ➤ Explain difference between native apps and HTML5 apps ➤ Describe advantages of native and HTML5 apps ➤ List steps to convert HTML5 apps to native apps	
18	HTML5 – TL18	All the topics as listed below from Session 14 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 14 – HTML5 Geolocation and APIs</u> ➤ Explain geolocation and its use in HTML5 ➤ Explain the Google Maps API ➤ Explain the drag-and-drop operations in HTML5 ➤ Explain the concept of Application Cache	<u>FrontEnd Technologies for Beginners</u> SG – Session 14 XP – Session 14 TG – Session 14
19	HTML5 – TL19	All the topics as listed below from Session 15 and Session 16 of <i>FrontEnd Technologies for Beginners</i> book should be covered in this session. <u>Session 15 – Building Mobile Web Applications</u> ➤ Describe features of different mobile devices ➤ List different types of platforms available for mobile devices ➤ Explain design and architectural aspects of a mobile Website ➤ Explain the requirements for developing and testing of a mobile Website ➤ Explain HTML5 support for a	<u>FrontEnd Technologies for Beginners</u> SG – Session 15 & 16 XP – Session 15 & 16 TG – Session 15 & 16

Session No.	Session Title	Session Details	Deliverables' Mapping
		mobile Website ➤ List the best practices for optimizing a mobile Website <u>Session 16 – Mini Project</u> ➤ Explain the use of HTML5, CSS, and JavaScript in a real-world application support for a mobile Website	
20	HTML5 – TL20	All the Workshop topics from Session 15 and Session 16 of <i>FrontEnd Technologies for Beginners</i> course should be covered in this session.	<u>FrontEnd Technologies for Beginners</u> Session 15 & 16
	S5-S6	All the Assignment topics from Session 15 and Session 16 of <i>FrontEnd Technologies for Beginners</i> course should be covered in this session.	<u>FrontEnd Technologies for Beginners (OnlineVarsity)</u> Session 15 & 16

5. Library References

➤	HTML5: Up and Running by Mark Pilgrim
➤	Introducing HTML5 (Voices That Matter) by Bruce Lawson
➤	Head First HTML5 Programming: Building Web Apps with JavaScript by Eric Freeman, Elisabeth Robson

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